

# External-Constraint Test (SCALULAR-classified) — Audit Ledger v1.0

**Test (canon form):** A boundary, symmetry, or equivalence claim PASSES only if it constrains a *family* of objects via a structural property *external* to the claim's own definition. If the claim is recovered just by unpacking the definition of the quantity it's about, it FAILS — it's a restatement wearing a name, not a finding.

**Scope of this pass:** every claim explicitly flagged, proposed, or tested across this session and the directly-linked prior sessions. Not an exhaustive sweep of the full ~2,078-instrument archive — flag if you want a deeper pass against a named instrument set.

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## PASS

| Claim                                                                      | External structural property invoked                                                                            |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| VEILED-Z                                                                   | Petermann factor divergence — derived from non-orthogonal mode coupling, not asserted                           |
| Funnel / branch-point cone geometry                                        | Universal cone geometry near a branch point — independent geometric fact                                        |
| Conditioned-RESPONSE_ZONE                                                  | Behavioral gate design grounded in stated mechanism, not label-matching                                         |
| Normality Exclusion Boundary (N(Z) Foreclosure Condition)                  | $SO(2)$ abelian closure — holds for the entire rotation family, proven independent of $\Delta(J)$ 's definition |
| CORDIC trajectory $\leftrightarrow$ eigenvector conjugate pair (Reading B) | Same conjugate symmetry $e^{\pm i\theta}$ appears at iteration level and operator level — derived, not assigned |

## FAIL — with repair action

| Claim                                                                               | Why it fails                                                                                        | Repair                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JAS×EMA=72 $\equiv$ 36×2<br>(DOB chain)                                             | Both independently constructed to hit 72; no external property forces them to be the <i>same</i> 72 | Retain 72 as a fixed canonical anchor (already locked). Remove " $\equiv$ " / convergence framing. State plainly: two independent constructions reaching the same target value — not evidence of a shared mechanism. |
| $\varphi \equiv$ EP branch exponent                                                 | Different universality classes (established prior session)                                          | Keep $\varphi$ and the EP branch exponent as separate, unequated constructs.                                                                                                                                         |
| $V(\psi) \equiv$ JAS(EMA)                                                           | Static object equated to dynamic object                                                             | Do not equate. Retain both, separately defined.                                                                                                                                                                      |
| "Defectiveness Exclusion Boundary" (EP-side mirror of Normality Exclusion Boundary) | Tautological — $\Delta(J) > 0$ is the definition of non-normal; nothing external is constrained     | Do not name. Refer to descriptively only ("non-normal by definition at the EP"), no boundary-name treatment.                                                                                                         |
| N(Z) / ZEUS as Yin-Yang, Reading A (the dial's two poles)                           | No mutual containment, no cyclical return — a static bounded interval, not Yin-Yang                 | Retain N(Z) as a bounded index. Drop Yin-Yang framing for the dial itself; reserve it for Reading B only.                                                                                                            |

## FLAGGED — not failed, held to lower evidentiary status

| Claim                              | Status                                                                                                                                                                                                          |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ZEUS "Zipper" term                 | Interpretive label for a real, derived mechanism (eigenvector convergence) — acceptable as naming, <b>not citable as independent proof</b> of that mechanism.                                                   |
| "United Spectrum" naming asymmetry | Names the $N = -1$ pole exactly (eigenvalue coalescence, derived); does not separately name $N = +1$ . Property of the <i>label</i> , not a defect in the index. No repair needed — flagged for awareness only. |

**Net read:** the test cleanly separates two classes that look superficially similar in this corpus — derivations using an external structural fact (group closure, divergence behavior, geometric universality) vs. constructions where two independently-chosen numbers or labels are pointed at each other and called convergence. Five pass, four fail, two flagged. The fails cluster on one signature every time: numerology/labeling without an external constraint doing the work.

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**Recursive Validity Check (self-application)**

The test can be applied to itself, using the deduction/induction cross rather than self-assertion:

| Arm       | Path                | Content                                                                                                                                                                                                                                 |
|-----------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deduction | Theory → Prediction | Theory = the analytic/synthetic distinction (standard, ERES-external epistemology). Prediction = the test's specific rule ("must constrain via an external structural property").                                                       |
| Induction | Data → Inference    | Data = the nine audited cases above, sorted independently. Inference = the pattern recovered from them: fails share an independently-constructed-numbers/labels signature; passes share a derivation-from-external-invariant signature. |

**Result:** the two arms converge on the same rule from independent directions — the test passes itself.

**Standing condition for any future recursive pass:** this only holds because the *Theory* node sits outside the ERES corpus (general epistemology, not an ERES-internal claim). If a future self-check used an ERES-internal claim as the Theory node, the check would fail by its own rule — two internally-constructed things validating each other is exactly the JAS×EMA=72≡36×2 failure mode, one level up. Any recursive application must re-verify this condition before counting as a pass.